

Due on monday, December 11th, in the morning class

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1. [4 pts.] Show that an operator, when expressed as a matrix in the basis of its own eigenstates, is diagonal.

Hint: Let's make this more specific: let $\{|w_n\rangle\}$ be the orthonormalized eigenstates of the operator $\hat{Y}$. Use the set $\{|w_n\rangle\}$ as basis set and express $\hat{Y}$ as a matrix in this representation. Show that this matrix is diagonal.

2. Consider a system of two spin-1/2 objects. (E.g., the spins of two ions trapped next to each other, or the electron and proton in a hydrogen atom, where we ignore spatial motion and concentrate on their spins.) The Hilbert space is 4-dimensional and is spanned by the basis

$$|\uparrow\uparrow\rangle \quad |\uparrow\downarrow\rangle \quad |\downarrow\uparrow\rangle \quad |\downarrow\downarrow\rangle$$

with the obvious meanings: e.g., $|\uparrow\downarrow\rangle$ represents the state where the first spin has z -component up ($+\hbar/2$) and the second spin has z -component down ($-\hbar/2$).

Using these four states as the basis set, a general state can be written as

$$|\psi\rangle = \alpha|\uparrow\uparrow\rangle + \beta|\uparrow\downarrow\rangle + \gamma|\downarrow\uparrow\rangle + \delta|\downarrow\downarrow\rangle = \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix}$$

- (a) [2 pts.] In the state $|\psi\rangle$, what is the probability of finding the first spin to have z -component up?
- (b) [2 pts.] What is the probability of finding the second spin to have z -component down?
- (c) [2 pts.] In a simultaneous measurement of the z -components of both the spins, one finds the first spin to be spin-up, and the second to be spin-down. What is the state of the system immediately after the measurement?
- (d) [2 pts.] Imagine instead a measurement of only the second spin. The result is found to be spin-up. What is the state of the system immediately after the measurement?

3. (a) [4 pts.] Consider the operator $\hat{A} = \frac{d}{dx}$. Which of the following functions are eigenfunctions of $\hat{A}$? (Show your calculation/argument.)

$$f_1(x) = e^{ikx} \quad f_2(x) = ax \quad f_3(x) = \cos(kx) \quad f_4(x) = e^{-ax^2}$$

Here k and a are real constants. For those functions that are eigenfunctions, give the eigenvalue.

- (b) [5 pts.] Consider the operator $\hat{B} = \frac{d^2}{dx^2}$. Which of the above functions are eigenfunctions of $\hat{B}$? For those functions that are eigenfunctions, give the eigenvalue.

4. The quantum harmonic oscillator.

We denote the orthonormalized energy eigenstates of the harmonic oscillator by $|n\rangle$, on which the creation/annihilation operators act as follows:

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle \quad \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

- (a) [7 pts.] Express $\hat{x}^2$ in terms of $\hat{a}$ and $\hat{a}^\dagger$. Hence show that the uncertainty of the position in the state $|n\rangle$ is

$$\Delta x = \sigma \sqrt{n + \frac{1}{2}}$$

- (b) [7 pts.] Express $\hat{p}^2$ in terms of $\hat{a}$ and $\hat{a}^\dagger$. Hence calculate the uncertainty of the momentum, Δp , in the state $|n\rangle$.
- (c) [3 pts.] Find the uncertainty product $\Delta x \Delta p$ in the state $|n\rangle$. Also express this quantity as a function of the eigenenergy E_n .

5. (a) [SELF] Show that for any three operators $\hat{A}$, $\hat{B}$, and $\hat{C}$,

$$[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$$

- (b) [6 pts.] Use the relation $[\hat{x}, \hat{p}] = i\hbar$ and the result above to prove by mathematical induction the identity

$$[\hat{x}^n, \hat{p}] = in\hbar\hat{x}^{n-1}$$

for any positive integer n .

(If unfamiliar, please look up “mathematical induction”, a standard technique for proving statements for all positive integers n .)

6. Consider the Schroedinger equation for a single particle in one dimension:

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x, t) + V(x) \psi(x, t)$$

The potential $V(x)$ is real-valued, i.e., $V^*(x) = V(x)$.

From the wavefunction one can obtain the probability density $\rho = |\psi(x, t)|^2 = \psi^*(x, t) \psi(x, t)$ and the probability current density

$$j = \frac{\hbar}{2mi} \left[\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right] \quad \text{omitting arguments } (x, t).$$

- (a) [**6 pts.**] Using the Schroedinger equation, show that ρ and j obey the one-dimensional continuity equation

$$\frac{\partial j}{\partial x} = -\frac{\partial \rho}{\partial t}$$

- (b) [**SELF**] Guess the form of the continuity equation for the probability density, for the case of a particle in three spatial dimensions.
- (c) [**SELF**] What does the continuity equation imply physically?
The wikipedia page on “Continuity Equation” has a section called “quantum mechanics”. You might try reading through.